



Silhouette Score (main metric)

Davies-Bouldin Index (secondary)

Calinski-Harabasz Index (secondary)

Runtime (computation time for each pipeline)

use when? before clustering or after clustering? how?

All clustering metrics (Silhouette, Davies-Bouldin, Calinski-Harabasz) are computed **AFTER clustering** – you need the **cluster labels** to calculate them.

When and How to Use Each Metric

Step-by-Step Pipeline:

1. Original Data (18-20 variables)
↓
2. Dimensionality Reduction (PCA or UMAP)
→ Reduced data (e.g., 2-10 dimensions)
↓
3. Clustering (KMeans, HDBSCAN, etc.)
→ Cluster labels assigned to each sample
↓
4. [COMPUTE METRICS HERE] ← Use cluster labels + reduced data
↓
5. Visualization (PCA 2D / UMAP 2D)
↓
6. Surrogate Model (RF, DT, XG)
→ Training time for runtime measurement

Clustering Metrics (After Clustering, Step 4)

All three metrics require:

- **Input 1:** Your reduced-dimensional data (after PCA/UMAP)
- **Input 2:** Your cluster labels (from clustering algorithm)

Python Code Example:

```
from sklearn.metrics import silhouette_score, davies_bouldin_score, calinski_harabasz_score

# After clustering:
# X_reduced = data after PCA/UMAP
# cluster_labels = output from KMeans/HDBSCAN/etc.

silhouette = silhouette_score(X_reduced, cluster_labels) # Higher = better
davies_bouldin = davies_bouldin_score(X_reduced, cluster_labels) # Lower = better
calinski_harabasz = calinski_harabasz_score(X_reduced, cluster_labels) # Higher = better
```

What Each Metric Measures:

Metric	Formula	Better	When to use
Silhouette	$(b - a) / \max(a, b)$	Higher (-1 to 1) [1] [2]	General-purpose; measures cohesion vs separation [1]
Davies-Bouldin	$\max(\text{mean_dist_i} / \text{dist_ij})$	Lower [1] [2]	Minimizing intra-cluster variance, maximizing inter-cluster separation [1]
Calinski-Harabasz	$\text{var_between} / \text{var_within} * (n-1) / (k-1)$	Higher [1] [2]	Favors compact, well-separated clusters; good for balanced clusters [1]

Runtime (Measure Twice)

Runtime should be measured for **both clustering step and surrogate model training**:

When to Measure:

```
import time

# 1. Clustering runtime
start = time.time()
cluster_labels = clustering_model.fit_predict(X_reduced)
clustering_time = time.time() - start

# 2. Surrogate model training runtime
start = time.time()
surrogate_model.fit(X_reduced, cluster_labels) # Training with cluster labels as "targets"
surrogate_training_time = time.time() - start

# Total pipeline runtime
total_runtime = clustering_time + surrogate_training_time
```

Summary Table

Metric	When to Compute	Inputs Needed	Better Score
Silhouette Score	After clustering	X_reduced + cluster_labels	Higher (-1 to 1) ^[1] _{[2] [3]}
Davies-Bouldin Index	After clustering	X_reduced + cluster_labels	Lower ^[1] ^[2] ^[3]
Calinski-Harabasz Index	After clustering	X_reduced + cluster_labels	Higher ^[1] ^[2] ^[3]
Runtime	During clustering + surrogate training	Time before/after fit_predict() and fit()	Lower

Key Takeaway

- All clustering metrics are computed **AFTER** clustering (Step 4) ^[1] ^[2] ^[3]
- They are **internal validation metrics** — no ground truth needed ^[2] ^[3]
- Use **cluster labels + reduced data** as inputs to all three metrics
- Measure **runtime** during the clustering and surrogate model steps

This gives you a complete evaluation framework for comparing your 6 pipelines!

✱✱

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4. <https://towardsdatascience.com/7-evaluation-metrics-for-clustering-algorithms-bdc537ff54d2/>
5. <https://codesignal.com/learn/courses/cluster-performance-unveiled/lessons/evaluating-k-means-clustering-performance-with-python-metrics>
6. <https://www.kaggle.com/code/misterfour/k-means-clustering-optimal-number-of-clusters>
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8. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10538130/>
9. <https://www.youtube.com/watch?v=MHnoWsBJpeM>
10. <https://www.mathworks.com/help/stats/evalclusters.html>